PYTHON MEGA INTERVIEW & PRACTICE QUESTION BANK

(For Data Analyst, Data Science & Software Roles)

Companies: TCS, Infosys, Wipro, Accenture, Cognizant, Capgemini, Deloitte, EY, PwC, IBM, Amazon, Google, Microsoft

Created By:

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1. Python History – 20 Questions

1. Who developed Python? – (TCS, Infosys, Wipro, Capgemini)
2. In which year was Python first released? – (Accenture, Cognizant, IBM, HCL)
3. Why is Python named "Python"? – (Google, Amazon, Microsoft, Meta)
4. Is Python an interpreted or compiled language? – (TCS, Wipro, Infosys, Tech Mahindra)
5. What is the full name of Python's creator? – (IBM, Cognizant, Capgemini, Accenture)
6. Is Python open-source? – (Amazon, Google, Meta, Oracle)
7. What is the official website of Python? – (Infosys, TCS, Wipro, Deloitte)
8. Which was the first stable version of Python? – (Accenture, Capgemini, Cognizant, EY)

9. What is Python's design philosophy? – (Google, Microsoft, Amazon, Meta)
10. What is the main goal of Python? – (IBM, Wipro, TCS, Infosys)
11. What is the Zen of Python? – (Google, Amazon, Meta, Netflix)
12. Why is Python considered beginner-friendly? – (TCS, Infosys, Wipro, Capgemini)
13. What are the major versions of Python? – (Accenture, Cognizant, Deloitte, EY)
14. Difference between Python 2 and Python 3? – (Wipro, Capgemini, IBM, Deloitte)
15. Why is Python popular in Data Science? – (Amazon, Google, Microsoft, Meta)
16. Why is Python called a scripting language? – (Infosys, Wipro, Capgemini, IBM)
17. Why is Python platform independent? – (TCS, Infosys, Wipro, Capgemini)
18. Why is Python dynamic typed? – (Accenture, Cognizant, Deloitte, EY)
19. What industries use Python the most? – (Amazon, Microsoft, Google, Meta)
20. Why is Python future-proof? – (Amazon, Microsoft, Google, Meta)

2. Python Version Checking – 20 Questions

1. How to check Python version? – (TCS, Infosys, Wipro, Capgemini)
2. Command to check Python version in terminal? – (Accenture, Cognizant, IBM, HCL)
3. How to check Python version inside a script? – (Amazon, Google, Microsoft, Meta)
4. How to check Python version using sys module? – (TCS, Infosys, Wipro, Deloitte)
5. How to check Python version in Jupyter Notebook? – (Accenture, Capgemini, Cognizant, EY)
6. How to check installed Python versions? – (Google, Amazon, Microsoft, Meta)
7. How to check Python architecture (32/64 bit)? – (IBM, Wipro, Infosys, HCL)
8. How to check Python version in virtual environment? – (Accenture, Cognizant, Deloitte, EY)
9. How to verify Python installation? – (Amazon, Google, Microsoft, Meta)
10. Difference between sys.version and sys.version_info? – (TCS, Infosys, Wipro, Capgemini)
11. How to check Python version in Linux? – (IBM, Wipro, Infosys, HCL)
12. How to check Python version in Windows? – (Accenture, Cognizant, Deloitte, EY)

13. How to check Python version in MacOS? – (Google, Amazon, Microsoft, Meta)
14. How to check default Python version? – (TCS, Infosys, Wipro, Capgemini)
15. How to change default Python version? – (Accenture, Cognizant, Deloitte, EY)
16. How to detect Python compatibility issues? – (IBM, Wipro, Infosys, HCL)
17. How to verify environment version? – (Amazon, Google, Microsoft, Meta)
18. How to update Python version safely? – (TCS, Infosys, Wipro, Deloitte)
19. How to downgrade Python version? – (Accenture, Capgemini, Cognizant, EY)
20. How to handle version mismatch errors? – (Google, Amazon, Microsoft, Meta)

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3. Variables – 20 Questions

1. What is a variable in Python? – (TCS, Infosys, Wipro, Capgemini)
2. Variable naming rules in Python? – (Accenture, Cognizant, IBM, HCL)
3. How to declare variables in Python? – (Amazon, Google, Microsoft, Meta)
4. What is dynamic typing? – (TCS, Infosys, Wipro, Deloitte)

5. Multiple variable assignment? – (Accenture, Capgemini, Cognizant, EY)
6. Swap two variables without third variable. – (Google, Amazon, Microsoft, Meta)
7. Difference between local and global variables. – (IBM, Wipro, TCS, Infosys)
8. What is constant in Python? – (Accenture, Cognizant, Deloitte, EY)
9. How to delete a variable? – (Amazon, Google, Microsoft, Meta)
10. What is variable scope? – (TCS, Infosys, Wipro, Capgemini)
11. What is nonlocal variable? – (Accenture, IBM, Cognizant, HCL)
12. What is identifier? – (Google, Amazon, Microsoft, Meta)
13. Keyword vs variable? – (TCS, Infosys, Wipro, Capgemini)
14. How to get variable address? – (Accenture, Cognizant, Deloitte, EY)
15. What is reference variable? – (IBM, Wipro, Infosys, HCL)
16. Variable naming best practices? – (Google, Amazon, Microsoft, Meta)

17. Can variable change its data type? – (TCS, Infosys, Wipro, Capgemini)
18. What is underscore variable? – (Accenture, Cognizant, Deloitte, EY)
19. What are private variables? – (IBM, Wipro, Infosys, HCL)
20. Real-time use of variables? – (Amazon, Google, Microsoft, Meta)

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4. Operators (All Types) – 20 Questions

1. What is an operator? – (TCS, Infosys, Wipro, Capgemini)
2. Types of operators in Python? – (Accenture, Cognizant, IBM, HCL)
3. Arithmetic operators? – (Amazon, Google, Microsoft, Meta)
4. Comparison operators? – (TCS, Infosys, Wipro, Deloitte)
5. Logical operators? – (Accenture, Capgemini, Cognizant, EY)
6. Assignment operators? – (Google, Amazon, Microsoft, Meta)
7. Bitwise operators? – (IBM, Wipro, Infosys, HCL)
8. Membership operators? – (Amazon, Google, Microsoft, Meta)
9. Identity operators? – (TCS, Infosys, Wipro, Capgemini)

10. Difference between == and is? – (Accenture, Cognizant, Deloitte, EY)
11. Use of ** operator? – (IBM, Wipro, Infosys, HCL)
12. Use of // operator? – (Amazon, Google, Microsoft, Meta)
13. Operator precedence? – (TCS, Infosys, Wipro, Capgemini)
14. Short-circuit evaluation? – (Accenture, Cognizant, Deloitte, EY)
15. XOR operator? – (IBM, Wipro, Infosys, HCL)
16. Bit shifting? – (Amazon, Google, Microsoft, Meta)
17. in vs not in? – (TCS, Infosys, Wipro, Capgemini)
18. += operator usage? – (Accenture, Cognizant, Deloitte, EY)
19. Real-time operator usage? – (IBM, Wipro, Infosys, HCL)
20. Operator best practices? – (Amazon, Google, Microsoft, Meta)

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5. Type & Type Conversion – 20 Questions

1. What is data type? – (TCS, Infosys, Wipro, Capgemini)
2. Built-in data types in Python? – (Accenture, Cognizant, IBM, HCL)

3. What is type casting? – (Amazon, Google, Microsoft, Meta)
4. Implicit type conversion? – (TCS, Infosys, Wipro, Deloitte)
5. Explicit type conversion? – (Accenture, Capgemini, Cognizant, EY)
6. int() usage? – (Google, Amazon, Microsoft, Meta)
7. float() usage? – (IBM, Wipro, Infosys, HCL)
8. str() usage? – (Amazon, Google, Microsoft, Meta)
9. bool() usage? – (TCS, Infosys, Wipro, Capgemini)
10. Convert string to int? – (Accenture, Cognizant, Deloitte, EY)
11. Convert list to tuple? – (IBM, Wipro, Infosys, HCL)
12. Convert tuple to list? – (Amazon, Google, Microsoft, Meta)
13. Convert dict to list? – (TCS, Infosys, Wipro, Capgemini)
14. Convert set to list? – (Accenture, Cognizant, Deloitte, EY)
15. Numeric conversions? – (IBM, Wipro, Infosys, HCL)
16. Binary conversion? – (Amazon, Google, Microsoft, Meta)
17. Type mismatch error? – (TCS, Infosys, Wipro, Capgemini)

18. Safe type conversion? – (Accenture, Cognizant, Deloitte, EY)
19. Type conversion risks? – (IBM, Wipro, Infosys, HCL)
20. Industry use of type casting? – (Amazon, Google, Microsoft, Meta)

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6. Conditional Statements – 20 Questions

1. What is if statement? – (TCS, Infosys, Wipro, Capgemini)
2. Syntax of if-else? – (Accenture, Cognizant, IBM, HCL)
3. if-elif-else usage? – (Amazon, Google, Microsoft, Meta)
4. Nested if? – (TCS, Infosys, Wipro, Deloitte)
5. Ternary operator? – (Accenture, Capgemini, Cognizant, EY)
6. Multiple conditions? – (Google, Amazon, Microsoft, Meta)
7. Boolean expressions? – (IBM, Wipro, Infosys, HCL)
8. Logical operators in if? – (Amazon, Google, Microsoft, Meta)
9. Indentation importance? – (TCS, Infosys, Wipro, Capgemini)
10. Switch alternative in Python? – (Accenture, Cognizant, Deloitte, EY)
11. Menu driven programs? – (IBM, Wipro, Infosys, HCL)

12. Decision making logic? – (Amazon, Google, Microsoft, Meta)
13. Nested condition? – (TCS, Infosys, Wipro, Deloitte)
14. Short-hand if? – (Accenture, Cognizant, Deloitte, EY)
15. Condition validation? – (IBM, Wipro, Infosys, HCL)
16. Error handling with if? – (Amazon, Google, Microsoft, Meta)
17. Performance impact? – (TCS, Infosys, Wipro, Capgemini)
18. Conditional filtering? – (Accenture, Cognizant, Deloitte, EY)
19. Data validation logic? – (IBM, Wipro, Infosys, HCL)
20. Industry usage? – (Amazon, Google, Microsoft, Meta)

. For Loop – 20 Questions

1. What is a for loop in Python? – (TCS, Infosys, Wipro, Capgemini)
2. What is the syntax of for loop? – (Accenture, Cognizant, IBM, HCL)
3. What is the use of range() function? – (Amazon, Google, Microsoft, Meta)
4. How does for loop work with lists? – (TCS, Infosys, Wipro, Deloitte)

5. How to iterate through a string using for loop? – (Accenture, Capgemini, Cognizant, EY)
6. What is nested for loop? – (Google, Amazon, Microsoft, Meta)
7. How to use break in for loop? – (IBM, Wipro, Infosys, HCL)
8. How to use continue in for loop? – (Amazon, Google, Microsoft, Meta)
9. What is enumerate() function? – (TCS, Infosys, Wipro, Capgemini)
10. What is zip() function? – (Accenture, Cognizant, Deloitte, EY)
11. What is for-else loop? – (IBM, Wipro, Infosys, HCL)
12. How to iterate through dictionary using for loop? – (Amazon, Google, Microsoft, Meta)
13. What is list comprehension? – (TCS, Infosys, Wipro, Deloitte)
14. Difference between for loop and while loop? – (Accenture, Capgemini, Cognizant, EY)
15. Reverse iteration using for loop? – (IBM, Wipro, Infosys, HCL)
16. How to loop through multiple lists together? – (Amazon, Google, Microsoft, Meta)
17. What is loop optimization? – (TCS, Infosys, Wipro, Capgemini)

18. How to get index and value together? – (Accenture, Cognizant, Deloitte, EY)
19. Real-world example of for loop? – (IBM, Wipro, Infosys, HCL)
20. Industry use of for loops? – (Amazon, Google, Microsoft, Meta)

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8. While Loop – 20 Questions

1. What is while loop in Python? – (TCS, Infosys, Wipro, Capgemini)
2. Syntax of while loop? – (Accenture, Cognizant, IBM, HCL)
3. Difference between while and for loop? – (Amazon, Google, Microsoft, Meta)
4. What is infinite while loop? – (TCS, Infosys, Wipro, Deloitte)
5. How to stop infinite loop? – (Accenture, Capgemini, Cognizant, EY)
6. What is break statement? – (Google, Amazon, Microsoft, Meta)
7. What is continue statement? – (IBM, Wipro, Infosys, HCL)
8. What is while-else? – (Amazon, Google, Microsoft, Meta)
9. Counter-controlled while loop? – (TCS, Infosys, Wipro, Capgemini)

10. Condition-controlled while loop? – (Accenture, Cognizant, Deloitte, EY)
11. Input-based while loop? – (IBM, Wipro, Infosys, HCL)
12. Nested while loop? – (Amazon, Google, Microsoft, Meta)
13. Do-while loop alternative in Python? – (TCS, Infosys, Wipro, Deloitte)
14. Menu-driven program using while loop? – (Accenture, Capgemini, Cognizant, EY)
15. Loop debugging techniques? – (IBM, Wipro, Infosys, HCL)
16. Performance issues in while loops? – (Amazon, Google, Microsoft, Meta)
17. Best practices for while loop? – (TCS, Infosys, Wipro, Capgemini)
18. Validation logic using while loop? – (Accenture, Cognizant, Deloitte, EY)
19. Data processing using while loop? – (IBM, Wipro, Infosys, HCL)
20. Industry usage of while loop? – (Amazon, Google, Microsoft, Meta)

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9. Indexing & Slicing – 20 Questions

1. What is indexing in Python? – (TCS, Infosys, Wipro, Capgemini)
2. What is positive indexing? – (Accenture, Cognizant, IBM, HCL)
3. What is negative indexing? – (Amazon, Google, Microsoft, Meta)
4. What is slicing? – (TCS, Infosys, Wipro, Deloitte)
5. Syntax of slicing? – (Accenture, Capgemini, Cognizant, EY)
6. What is start:end:step? – (Google, Amazon, Microsoft, Meta)
7. Reverse slicing? – (IBM, Wipro, Infosys, HCL)
8. List slicing? – (Amazon, Google, Microsoft, Meta)
9. String slicing? – (TCS, Infosys, Wipro, Capgemini)
10. Tuple slicing? – (Accenture, Cognizant, Deloitte, EY)
11. Multi-dimensional indexing? – (IBM, Wipro, Infosys, HCL)
12. Boolean indexing? – (Amazon, Google, Microsoft, Meta)
13. Numpy array indexing? – (TCS, Infosys, Wipro, Deloitte)
14. Pandas dataframe indexing? – (Accenture, Capgemini, Cognizant, EY)
15. Fancy indexing? – (IBM, Wipro, Infosys, HCL)

16. Slice assignment? – (Amazon, Google, Microsoft, Meta)
17. Large data slicing? – (TCS, Infosys, Wipro, Capgemini)
18. Performance impact of slicing? – (Accenture, Cognizant, Deloitte, EY)
19. Error handling in indexing? – (IBM, Wipro, Infosys, HCL)
20. Industry usage of slicing? – (Amazon, Google, Microsoft, Meta)

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10. Type Checking & Value Extraction – 20 Questions

1. What does type() function do? – (TCS, Infosys, Wipro, Capgemini)
2. What is isinstance() function? – (Accenture, Cognizant, IBM, HCL)
3. How to check multiple data types? – (Amazon, Google, Microsoft, Meta)
4. How to extract value from list? – (TCS, Infosys, Wipro, Deloitte)
5. How to extract value from tuple? – (Accenture, Capgemini, Cognizant, EY)
6. How to extract value from dictionary? – (Google, Amazon, Microsoft, Meta)

7. How to extract nested values? – (IBM, Wipro, Infosys, HCL)
8. How to extract values from JSON? – (Amazon, Google, Microsoft, Meta)
9. How to extract column from Pandas dataframe? – (TCS, Infosys, Wipro, Capgemini)
10. How to extract row from Pandas dataframe? – (Accenture, Cognizant, Deloitte, EY)
11. Boolean filtering in dataframe? – (IBM, Wipro, Infosys, HCL)
12. Multiple condition filtering? – (Amazon, Google, Microsoft, Meta)
13. Safe extraction techniques? – (TCS, Infosys, Wipro, Capgemini)
14. Extract data from API response? – (Accenture, Cognizant, Deloitte, EY)
15. Extract data from files? – (IBM, Wipro, Infosys, HCL)
16. Error-free extraction methods? – (Amazon, Google, Microsoft, Meta)
17. Extract values using loops? – (TCS, Infosys, Wipro, Capgemini)
18. Real-time data extraction example? – (Accenture, Cognizant, Deloitte, EY)
19. Performance optimized extraction? – (IBM, Wipro, Infosys, HCL)

20. Industry data extraction use cases? – (Amazon, Google, Microsoft, Meta)